University of Arkansas

CSCE 46103/56103 Artificial Intelligence

Fall 2026

Sample Exam

October 15, 2026 | 3:30 PM–4:45 PM | 75 minutes

Name: _____

UARK Email: _____

Course: ☐ CSCE 46103 (undergraduate) ☐ CSCE 56103 (graduate)

Exam information

- This is an open-book exam. You may bring your own paper materials, including handwritten or printed notes.
- No electronic devices are permitted.

Student instructions

Write your name, UARK Email, and course above. Use the spaces provided for your answers and supporting work. Show intermediate steps for calculations and explain your reasoning when requested.

Score record (instructor use)

Question	Earned	Available
Question 1		40
Question 2		60
Total		**100**

1. (40 points) A robot moves in a known, fixed grid with $N = 20$ traversable cells. Each legal move goes one cell north, south, east, or west and costs 1; revisiting cells is allowed. There are $k = 4$ dots on distinct cells, none at the start. Entering a dot's cell collects it permanently. The map and original dot locations are fixed problem data.

Compare two tasks in this same world:

- **Task A:** Reach a specified destination cell t; collecting dots is irrelevant.
- **Task B:** Collect all four dots; the robot may finish in any cell.

(a) (8 points) For Task A, specify what information one search state must store and define each variable you use. Give the condition for recognizing a goal state and an upper bound on the number of states. Explain why previously collected dots need not be recorded.

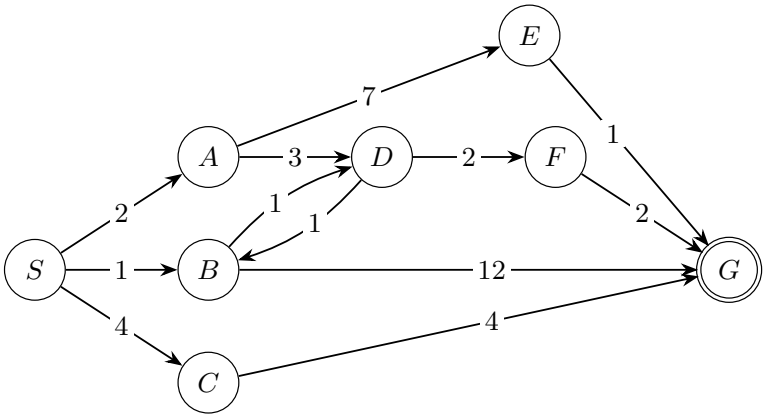
(b) (12 points) For Task B, define each variable in one search state and give its initial value, using symbols for the starting cell and initial dot locations. Describe how the state changes after a legal move, including a move onto a dot. State the condition for having collected all four dots.

(c) (6 points) Check whether position alone is sufficient. Describe two possible histories that leave the robot at the same cell but with different dots remaining. Explain why these situations must be distinguished for Task B. You may choose any connected grid consistent with the setup.

(d) (14 points) Count states and decide what to omit. For Task B, derive an upper bound in terms of N and k, then evaluate it for $N = 20$ and $k = 4$. (8 points)

A sensor also reports one of four facing directions and one of five battery levels. Neither affects legal actions, their results, their costs, or the goal in this model. By what factor would recording both enlarge the representation's upper bound? Should either be included? Explain. (6 points)

2. (60 points) Apply breadth-first search (BFS), depth-first search (DFS), and uniform-cost search (UCS) to the directed graph in Figure 1, from S to its only goal, G. Use the successor table's order; $A(2)$ means an edge to A with cost 2. Only the listed edges exist.



State	Ordered successors
S	$A(2), B(1), C(4)$
A	$D(3), E(7)$
B	$D(1), G(12)$
C	$G(4)$
D	$F(2), B(1)$
E	$G(1)$
F	$G(2)$
G	None

Figure 1: The search graph and its ordered successors. S is the start; the double circle marks goal G.

Begin each run with reached set $\{S\}$ and use these rules for the traces:

- DFS selects the first listed unreached successor next. UCS breaks equal-cost ties by earlier insertion.
- BFS and DFS mark states reached on generation, including states still in the frontier. Discard later paths to reached states and retain the first parent.
- BFS stops when the goal is generated, before enqueuing it. DFS and UCS stop when the goal is selected, before expanding it.
- UCS replaces a frontier entry's cost and parent only for a strictly cheaper path. Discard equal or more expensive paths and successors already selected.

How to use the trace tables: After each selection, show the resulting frontier with the next state on the left. Write new or changed parent assignments as $v \leftarrow u$ (u is v's parent). Write **STOP** on the stopping row; leave unused rows blank.

(a) (6 points) Prepare to search. Complete the table before tracing. Depth is the number of edges from S; g is their total cost.

Algorithm	What determines selection?	When is the goal tested?
BFS		
DFS		
UCS		

Question 2 continued: BFS

(b) (12 points) Trace BFS in the table below.

Step	Selected	Queue after step	New parents
0	—	$[S]$	S : none
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

When B proposes D, has D already been reached? Explain whether its parent changes.

Follow parents backward from G to S, then reverse that chain.

Returned path: _____

Number of edges: _____ Total cost (show the sum): _____

Question 2 continued: DFS

(c) (12 points) Trace DFS in the table below.

Step	Selected	Stack after step	New parents
0	—	$[S]$	S : none
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

When D proposes B, should another B enter the stack? Explain.

Use the parent chain to recover the solution.

Returned path: _____

Number of edges: _____ Total cost (show the sum): _____

Question 2 continued: UCS

(d) (18 points) Trace UCS in the table below. Write entries as $v : g(v)$, for example $S : 0$.

Step	Selected	Priority queue after step	Parent updates
0	—	$[S : 0]$	S : none
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

Check the updates to G explicitly. Fill in the stored cost just before each edge is considered, the candidate cost (show its sum), and the resulting parent of G.

Edge considered	Stored cost of G	Candidate cost	Parent afterward
$C \to G$			
$F \to G$			

Follow the final parent pointers to recover the solution.

Returned path: _____

Number of edges: _____ Total cost (show the sum): _____

Question 2 continued: comparison

(e) (12 points) Compare the answers and explain why. Copy the results of your traces into the table.

Algorithm	Returned path	Number of edges	Total cost
BFS			
DFS			
UCS			

Which result has the fewest edges? Which has the lowest cost? Explain why these can differ and connect each result to the algorithm's selection rule. (8 points)

If every edge instead cost 1, which of BFS, DFS, and UCS would be guaranteed to return a minimum-cost solution on this finite graph? Explain. (4 points)

Additional working space